XY160D Dual H-Bridge DC Motor Controller

Usage Guide

1. Specifications

Motor supply voltage	6.5 - 27V (24V rated)
Current	~7A continuous per channel, 16A peak
Channels	2 (drives two DC motors independently)
Control	4 signal pins (2 per motor)

2. Wiring

Pin	Connect to
VCC / +12V	Motor power supply (6.5 - 27V)
GND	Common ground (shared with MCU)
IN1, IN2	Motor A control (from Arduino/ESP GPIO)
IN3, IN4	Motor B control (from Arduino/ESP GPIO)
Motor A terminals	DC motor A
Motor B terminals	DC motor B

Important: Connect the GND of the motor supply and the MCU together.

3. Control Logic (per motor)

IN1	IN2	Result
HIGH	LOW	Forward
LOW	HIGH	Reverse
LOW	LOW	Stop (coast)
HIGH	HIGH	Brake

Same pattern applies for IN3/IN4 controlling Motor B.

4. Speed Control

Send a PWM signal to one of the two input pins instead of a plain HIGH. For example, with Arduino:

```
// Motor A on pins 5 (PWM) and 6 (PWM)
analogWrite(5, 200); // 0-255 speed
digitalWrite(6, LOW); // direction
```

5. Tips

- Add a capacitor (100uF+) across the motor power terminals to absorb voltage spikes.
- Do not power motors from the Arduino 5V - use a separate supply through the module VCC.
- If the motor spins the wrong way, swap IN1/IN2 in software or swap the motor wires.
- Ensure your motors are rated for the voltage you supply.